

 quarter = 4 equal size pieces.

1 litre = 1.06 ^{cuarto} quarts

4 quarts = 1 gallon

rounded to 108th how many litres per 1 gallon.

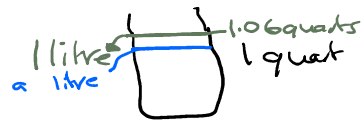
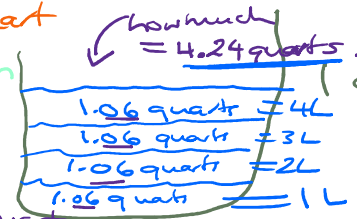
0.943 litre = 1 quart

3.77 litre = gallon

4 quarts = 1 gallon

"a" litres = 1 quart
1 litre = 1.06 quarts

$$a = \frac{1}{1.06} = 0.943$$



$$\begin{aligned} 1 \text{ litre} &= 1.06 \text{ quarts} \\ a \text{ litres} &= \frac{1}{1.06} \\ &= 0.943 \text{ litres} \end{aligned}$$

in 1 quart
4 quarts = 1 gallon

$$\begin{aligned} 4 \text{ litres} &= 4 \times 0.943 \\ &= 3.77 \text{ litres (gallon)} \end{aligned}$$

To make a drink

6-8 fl oz



6-8 fl oz per pouch mix

3 cups of water to 4 pouches of mix

1 cup = 8 fl oz
1 cup = 8 fl oz
1 cup = 8 fl oz
24 fl oz
4 pouches of mix

how much water do I need for 4 pouches?

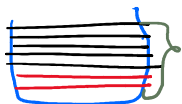
1 cup = 8 fl oz

fl = fluid
= liquid
oz = measure
= volume.

Unit	Sample Use	Comparing Units
fluid ounce (fl oz)	 1 fl oz	1 fl oz = $\frac{1}{8}$ c
cup (c)	 1 cup 8 fl oz	1 c = 8 fl oz
pint (pt)	 1 pt	1 pt = 2 c
quart (qt)	 1 qt	1 qt = 2 pt
gallon (gal)	 1 gal	1 gal = 4 qt

$$\begin{aligned} 24 \text{ fl oz of water} &= \frac{6 \text{ fl oz}}{\text{pouch of mix}} \\ 4 \text{ pouches of mix} & \end{aligned}$$

$$1 \text{ gallon} = 3.77$$



oil : vinegar

2 gallons = 5 litres

6 gallons = 15 litres
22.8 L = 15 L

3.8 litres = 1 gallon

22.64 litres

6 gallons (oil)

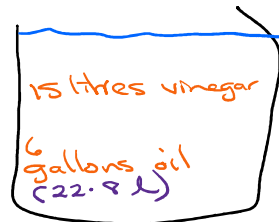
3 x amount

$$\begin{aligned} \text{Total litres} &= 15 + 22.8 \\ &= 37.8 \text{ litres} \end{aligned}$$

oil

Vinegar

2 gallons = 5 litres



x3

Afghanistan 2025 Page 2

90% or $\times 0.9 \Rightarrow 150 \times 0.9 = 135$

$$\begin{array}{r} x-3 \\ 2 \overline{) 2x-6} \\ \underline{2x-6} \\ 0 \end{array}$$

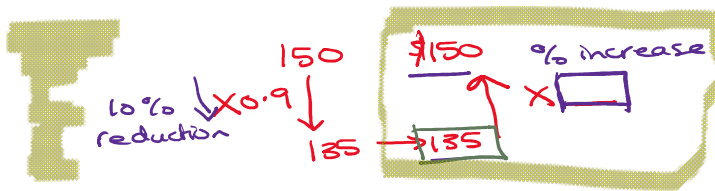
$$\begin{array}{r} x+3 \\ 3 \overline{) 3x+6} \\ \underline{3x+6} \\ 0 \end{array}$$

$$\begin{aligned} 2(x-3) + 4 &= 3(x+2) - 5 \\ 2x - 6 + 4 &= 3x + 6 - 5 \\ 2x - 2 &= 3x + 1 \\ -2 &= x + 1 \\ -3 &= x \\ x &= -3 \end{aligned}$$

Expand (get rid of the brackets)

$$\begin{aligned} 2x - 2 &= 3x + 1 \\ 2x - 2 &= 3x + 1 \\ -2 &= 3x + 1 \\ -3 &= x \\ -3 &= x \end{aligned}$$

Shop sells \$150, 10% off everything
new price = \$135 (150×0.9 or $150 - 15$)
(90%) 100% - 10%



10% of 135 is 13.50

$135 + 13.5 = 148.5$
Not \$150 that I started with!!!
 $\frac{10}{100} = \frac{10}{100} = 10\%$
 $0.9 = 0.90 = 90\%$

\$150 $\rightarrow$ 10% was \$15

So 10% discount = $150 - 15 = 135$

90% of 150 is $150 \times 0.9 = 135$

\$135 what is the % increase to get back to \$150

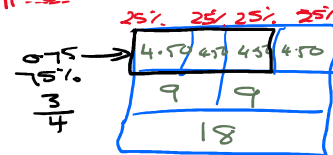
$135 \times \frac{0.112}{100} = 150.12$

1.111 actually mean

$135 \times \frac{1.111}{100} = \frac{150}{135}$
 $\frac{x}{100} = \frac{150}{135}$
 $= 1.111$

$\frac{11.11}{100} = 11.11\%$

Jacket on Sale! 25% off
original \$18 sale price.



calculate 25% & subtract

$18 \times \frac{25}{100} = 4.50$

$18 - 4.50 = 13.50$

multiply to take away 25%

$18 \times 0.75 = 13.50$

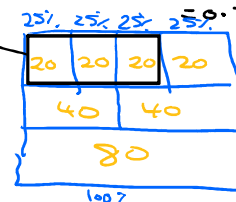
$\frac{75}{100} = \frac{3}{4}$

$\frac{75}{100} = \frac{3}{4}$

$\frac{75}{100} = \frac{3}{4}$

Pants on Sale 25% off
original price ~ \$80
sale price = ?

25% of 80
discount = \$20
sale price = $80 - 20 = 60$
 $80 \times \frac{75}{100} = 80 \times 0.75 = 60$



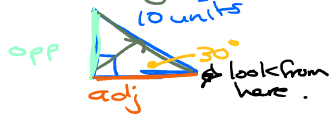
sale price = ?

$$\text{sale price} = \frac{80}{100} = 0.8$$

$$80 \times \frac{75}{100} = 80 \times 0.75 = 60$$

$$\frac{80}{100} = 0.8$$

side opposite the right angle
hypotenuse



opposite = across from
adjacent = next to

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

To find the opposite

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$10 \times \sin(30) = \frac{\text{opp}}{10} \times 10$$

$$10 \sin(30) = \text{opp}$$

$$\text{opp} = 5$$

$$\left. \begin{array}{l} \text{opp} = ? \\ \theta = 30 \\ \text{hyp} = 10 \end{array} \right\}$$

To find the adjacent

$$\cos(\theta) = \frac{\text{adj}}{\text{hyp}}$$

$$\cos(30) = \frac{\text{adj}}{10}$$

$$10 \times \cos(30) = \frac{\text{adj}}{10} \times 10$$

$$\begin{aligned} \text{adj} &= 10 \times \cos(30) \\ &= 5\sqrt{3} \text{ (Exact)} \\ &= 8.66 \text{ (Approx)} \end{aligned}$$

$$\left. \begin{array}{l} \text{adj} = ? \\ \text{hyp} = 10 \\ \theta = 30 \end{array} \right\}$$

You can make only four different cuboids with 16 cubes.

	Dimensions
Cuboid A	1 1 16
Cuboid B	1 2 8
Cuboid C	1 4 4
Cuboid D	2 2 4

(a) Which of the cuboids A and D has the larger surface area?

Tick (✓) the correct answer below.

- ☐ Cuboid A
☐ Cuboid D
☐ Both the same

Explain how you know.

(b) Which cuboid has the largest volume?

Tick (✓) the correct answer below.

- ☐ Cuboid A
☐ Cuboid B
☐ Cuboid C
☐ Cuboid D
☐ All the same

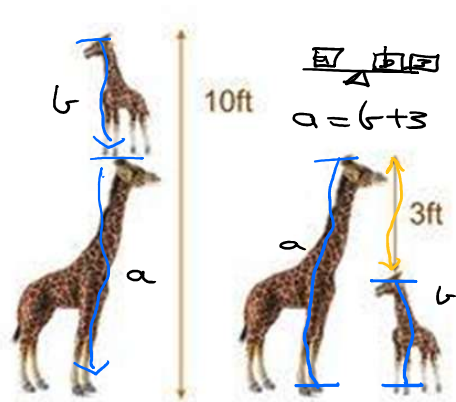
(c) How many of cuboid D make a cube of dimensions 4 x 4 x 4?

(d) You can make only six different cuboids with 24 cubes.

Complete the table to show the dimensions.

Two have been done for you.

	Dimensions
Cuboid E	1 1 24
Cuboid F	1 2 12
Cuboid G	
Cuboid H	
Cuboid I	



find height of each giraffe?

$a = \text{height of the adult giraffe}$
 $b = \text{height of the baby giraffe}$

$$a = b + 3$$

$$a + b = 10$$

$$a - b = 3$$

$$a + a = 2a$$

$$a \times a = a^2$$



$$a + b = 10 \quad (1)$$

$$a - b = 3 \quad (2)$$



Left
LHS

$$a + a = 2a$$

$$RHS \quad 2a = 13$$

$$a = \frac{13}{2}$$

$$a = 6.5$$

$$\frac{2a}{2} = \frac{13}{2} \quad (\div 2)$$

$$a = 6.5$$

31/1/25

$$a = 6.5$$

$$6.5 + b = 10$$

$$b = 10 - 6.5$$

$$b = 3.5$$

$$6.5 - b = 3$$

$$6.5 - 3.5 = b + 3 - 3$$

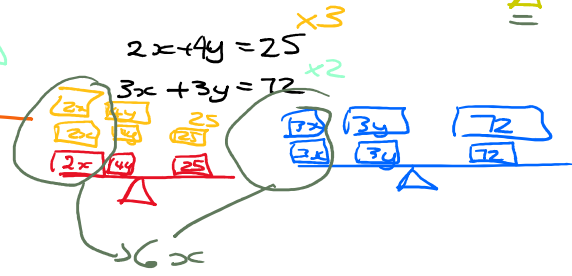
$$3.5 = b$$

$$2x + 4y = 25$$

$$3x + 3y = 72$$



cat



$$6x + 12y = 75$$

$$6x + 6y = 144$$

If a 13 ft board is cut into 3 equal pieces.
 How long is each piece?



Estimate/guess

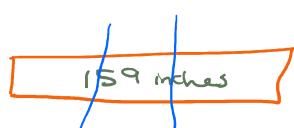
Answer is definitely more than $\frac{4 \text{ ft}}{3 \times 4 = 12}$
 less than $\frac{5 \text{ ft}}{3 \times 5 = 15}$.

13 ft 3 inches $\Rightarrow$ all inches.

1 foot = 12 inches.

$$13 \times 12 \text{ inches} + 3 \text{ inches} = 156 + 3$$

$$= 159 \text{ inches}$$



$$\frac{159}{3} = 53$$

so each piece is 53 inches

$$53 \text{ inches} = 4 \text{ ft} + 5 \text{ inches.}$$

$$4 \times 12 + 5$$

$$48 + 5$$

Learning is not event, it is a journey

Spain $\Rightarrow$ wonderful maths museum

Stone soup

bean shoot
(mung bean) $\rightarrow$ 1 week

oak tree $\rightarrow$ more than your
acorn lifetime



Patterns aka Algebra

Draw the next pattern in the sequence

Pattern Number (p)

Number of matches in each pattern

Rule m =

Number of matches in pattern 10

Pattern 1: $3 \times 1 + 1 = 4$

Pattern 2: $3 \times 2 + 1 = 7$

Pattern 3: $3 \times 3 + 1 = 10$

Pattern 4: $3 \times 4 + 1 = 13$

Matchsticks (m)

Pattern 5: $\text{matchsticks} = 3 \times \text{pattern} + 1$

1 4 7 10 13 16

Draw the next pattern in the sequence

Pattern Number (p)

1 2 3 4 5

Pattern 1: $3 \times 1 + 2 = 5$

Pattern 2: $3 \times 2 + 2 = 8$

Pattern 3: $3 \times 3 + 2 = 11$

Pattern 4: $3 \times 4 + 2 = 14$

Matchsticks (m)

Pattern 5: $\text{matchsticks} = 3 \times \text{pattern} + 2$

Draw the next pattern in the sequence

Pattern Number (p)

1 2 3 4 5

Pattern 1: $3 \times 1 + 1 = 4$

Pattern 2: $3 \times 2 + 1 = 7$

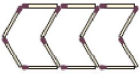
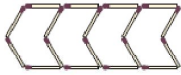
Pattern 3: $3 \times 3 + 1 = 10$

Pattern 4: $3 \times 4 + 1 = 13$

Matchsticks (m)

Pattern 5: $\text{matchsticks} = 3 \times \text{pattern} + 1$

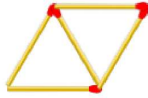
Patterns		Number of matches in each pattern					Rule $m =$	Number of matches in pattern 10
Draw the next pattern in the sequence		Pattern Number (p)						
    Pattern 1 Pattern 2 Pattern 3 Pattern 4 Matchsticks (m)								
Pattern 5								
Draw the next pattern in the sequence		Pattern Number (p)						
    Pattern 1 Pattern 2 Pattern 3 Pattern 4 Matchsticks (m)								
Pattern 5								

		Number of matches in each pattern					Rule $m =$	Number of matches in pattern 10
Draw the next pattern in the sequence		Pattern Number (p)						
 Pattern 1		 Pattern 2		 Pattern 3		 Pattern 4		
Pattern 5 								
Matchsticks (m)								

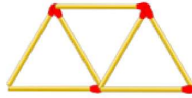
Question 1:



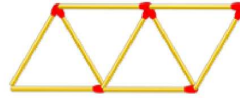
Pattern 1



Pattern 2



Pattern 3



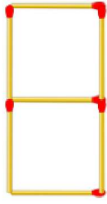
Pattern 4

Pattern Number:	1	2	3	4	...	n	...	100	
Number of Matchsticks:									87

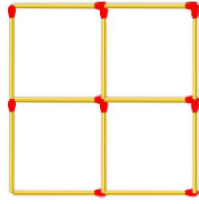
Rule in words: number of matchsticks = _____

Rule in symbols: $m =$ _____

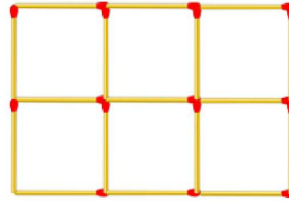
Question 2:



Pattern 1



Pattern 2



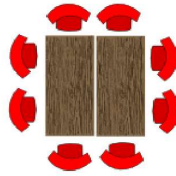
Pattern 3

Pattern Number:	1	2	3	4	...	n	...	114	
Number of Matchsticks:									122

Rule in words: number of matchsticks = _____

Rule in symbols: $m =$ _____

Question 3:

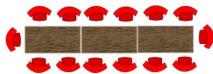
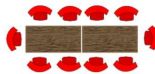


Number of Tables:	1	2	3	4	...	n	...	138
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 4:

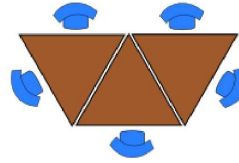
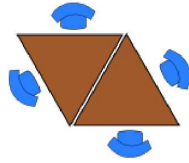


Number of Tables:	1	2	3	4	...	n	...	90
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 5:

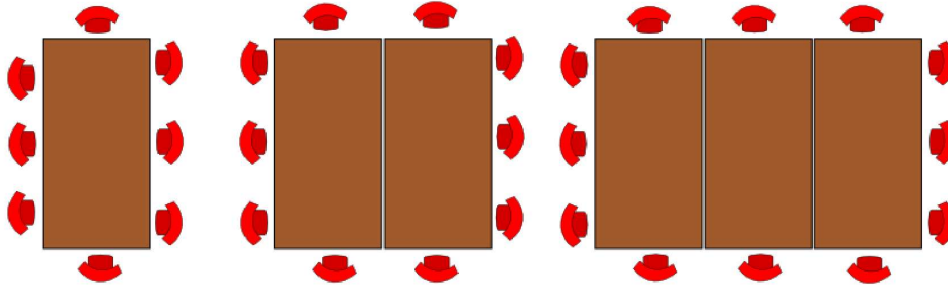


Number of Tables:	1	2	3	4	...	n	...	102
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 6:

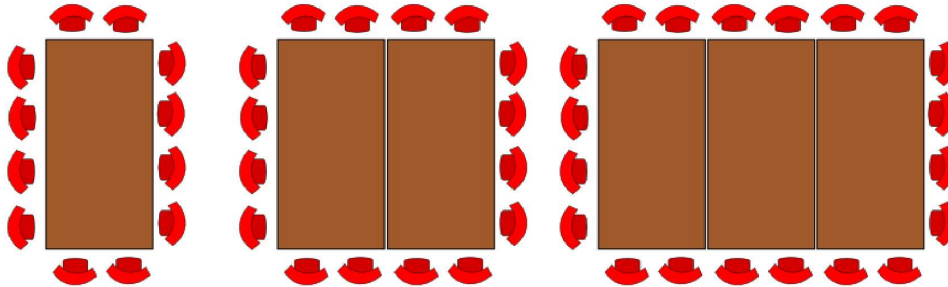


Number of Tables:	1	2	3	4	...	n	...	112
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 7:



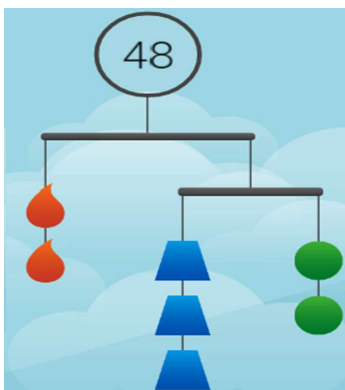
Number of Tables:	1	2	3	4	...	n	...	150	
Number of Chairs:									

In the space below, create your own (random!) pattern and ask another student to work out the rule for the pattern:



- A full jar of honey weighs 1000 grams.
- With half of the honey, the total weight is 600 grams.

What is the weight of the empty jar in grams?





walk at 7 km/h how long does

$$\star \text{ speed} = \frac{\text{distance}}{\text{time}} \quad \blacktriangle \text{ time} = \frac{\text{distance}}{\text{speed}}$$

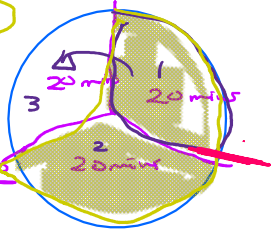
km/h? distance = speed $\times$ time.

km per hour? 1 Km 40 mins speed 1 km = 1000 m

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{1 \text{ Km}}{40 \text{ min}} = \frac{1000 \text{ m}}{40 \text{ min}} = 25 \text{ m/min}$$

$\frac{1}{2} = 20 \text{ mins.}$ km/hr 1 km 40 mins = $\frac{40}{60} = \frac{2}{3} \text{ hr}$ 60 mins = 1 hour

yellow = 1
 $\frac{1}{2} = \frac{1}{2}$ whole circle
 $\frac{2}{3}$

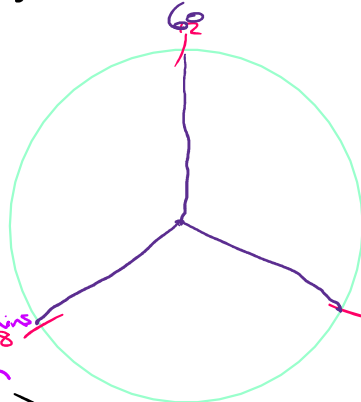
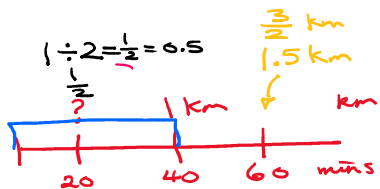


$$\text{speed} = \frac{1 \text{ km}}{\frac{2}{3} \text{ hr}}$$

$$1 \div \frac{2}{3} = 1 \times \frac{3}{2} = \frac{3}{2} \frac{\text{km}}{\text{hr}} = 1.5 \text{ km/hr.}$$

6 km $\Rightarrow$ takes 2 hrs = 3 km/hr

3 km = 1 hr
 $\frac{3 \text{ km}}{60 \text{ min}} = 2 \text{ hrs}$



how far did I run in 20 mins

$$\frac{40 \text{ mins}}{60 \text{ mins/hr}} = \frac{2}{3} \text{ hr}$$

1 km = 40 mins.

30 mins = $\frac{30}{60} = \frac{1}{2} \text{ hr} = 0.5 \text{ hr}$

180 mins = $\frac{180}{60} = 3 \text{ hours}$

60 60 60 = 3 hrs

200 mins = _____ hours.

$\frac{200}{60} = 3.3 \text{ hr} = 3 \text{ hr } 20 \text{ mins}$

150 mins = 2.5 hr
 60 60 60 = 3 hrs
 1 hr 1 hr 1 hr
 20 mins = $\frac{20}{60} = \frac{1}{3} \text{ hr} = 0.33$

length time
 $\frac{100}{60}$

$$60 \text{ seconds} = 1 \text{ min}$$

$$150 \text{ seconds} = 150 \times \frac{1}{60} = \frac{150}{60} = 2.5 \text{ mins}$$



$$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ 3 \text{ lb } 4 \text{ oz} \\ \hline 7 \text{ lb } 14 \text{ oz} \end{array} \quad \begin{array}{r} 1 \text{ lb } = ? \text{ oz} \\ 16 \text{ oz} \\ 3 \text{ lb } 7 \text{ oz} \\ \hline 7 \text{ lb } 17 \text{ oz} \\ = 8 \text{ lb } 1 \text{ oz} \end{array}$$

A candy store currently sells almonds for \$8.50 per kilogram and peanuts for \$4.50 per kilogram. It wants to create 20 kilograms of mixed nuts that will equivalently sell for \$5.30 per kilogram. How many kilograms of almonds should they use?



Algebra: Almonds \$8.50/kg, Peanuts \$4.50/kg, 20 kg bag sells for \$5.30/kg. $x = \text{no of kgs of Almonds}$, $y = \text{no of kgs of peanuts}$.

Guess & Check

Kg Almonds	Kg of Peanuts	\$ of nuts
10	10	$10 \times 8.50 + 10 \times 4.50 = 130$

$x = \text{no of kilograms of Almonds}$ (\$8.50/kg)
 $y = \text{no of kilograms of Peanuts}$ (\$4.50/kg)
 20 kg bag @ \$5.30/kg

No of kilograms of nuts: $x + y = 20$ (1)
 Cost of nuts: $8.5x + 4.5y = 106$ (2) ($106 = 5.30 \times 20$)

re-written (1): $y = 20 - x$ (3)
 sub (3) into (2): $8.5x + 4.5(20 - x) = 106$
 $8.5x + 90 - 4.5x = 106$

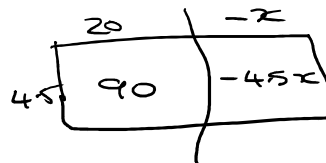
$$\frac{4x}{4} = \frac{16}{4}$$

$x = 4 \Rightarrow \text{No of kilograms of Almonds}$.

Almonds: 4 kg, Peanuts: 16 kg, 20 kg @ \$5.30/kg.

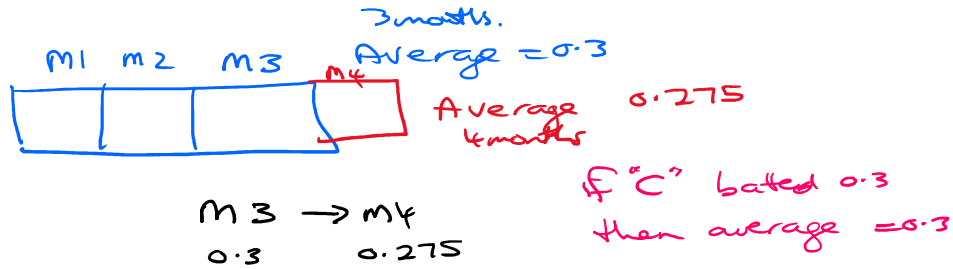
$$4 \times 8.5 + 16 \times 4.50 = \$106$$

34 + 72



Q. After the first three months of the season, a baseball player's batting average is .300. After another month, his average for the four months is .275. Which of the following is most likely? A. He did not play the last month. B. He batted about .200 the last month. C. He batted about .300 the last month. D. He batted about .350 the last month.

From <<https://meet.google.com/oap-ffqh-yrp>>



$$3 \times \frac{m_1 + m_2 + m_3}{3} = 0.3 \times 3$$

$$m_1 + m_2 + m_3 = 3 \times 0.3$$

$$\frac{m_1 + m_2 + m_3 + m_4}{4} = 0.3$$

$$\frac{0.9 + m_4}{4} = 0.3$$

option A: he did not play (scored 0)

$$0.9 + m_4 = 0.3 \times 4$$

$$0.9 + m_4 = 1.2$$

$$m_4 = 0.3$$

$$\frac{m_1 + m_2 + m_3}{3} = 0.3 \rightarrow \frac{0.9 + m_4}{4} = \text{average for 4 months}$$

$$m_1 + m_2 + m_3 = 3 \times 0.3 = 0.9$$

$$= \frac{0.9 + 0}{4} = 0.225$$

✓ Option B

$$\frac{m_1 + m_2 + m_3}{3} = 0.3$$

$$m_1 + m_2 + m_3 = 0.9$$

$$\frac{0.9 + m_4}{4} = 0.275$$

$$\frac{0.9 + 0.2}{4} = \frac{1.1}{4} = 0.275$$

Q. On the first day of a sale at Runway thing, all jeans were \$25 and all shirts were \$22. The store took in \$686. On the second day, the store reduced the price of both jeans and shirts to \$20 each. It sold exactly the same number of each item as it had on the first day. If the store made \$580 on the second day, how many shirts did they sell in all during the two-day sale?

From <<https://meet.google.com/oap-ffqh-yrp>>

$x = \text{no of jeans sold during the sale}$
 $y = \text{no of shirts sold during the sale.}$

first day

$$25x + 22y = 686 \quad (1)$$

2nd day

$$20x + 20y = 580 \quad (2)$$

$$\frac{25}{11}x + 2y = \frac{686}{11} \quad (3)$$

$$2x + 2y = 58 \quad (4)$$

$$(1) \div 11$$

$$(3) \div 10$$

$$(3) - (4)$$

$$\frac{25}{11}x - 2x = \frac{686}{11} - 58$$

$$\frac{25x - 22x}{11} = \frac{686 - 638}{11}$$

$$\frac{3x}{11} = \frac{48}{11}$$

$$3x = 48$$

$$x = 16$$

$$1^{\text{st}} \text{ option } (1) \div 5$$

$$(2) \div 4$$

so that the co-efficient of x is 1

$$2^{\text{nd}} \text{ option } (1) \div 11$$

$$(2) \div 20$$

everything $2y$

$$\frac{25x}{22} + y = \frac{686}{22}$$

$$25x + 22y = 686$$

③ + ④

$$\frac{25}{11}x - 2x = \frac{686}{11} \quad \text{SF}$$

$$\frac{25x - 22x}{11} = \frac{686}{11}$$

$$\cancel{11} \times \frac{3x}{11} = \frac{686}{\cancel{11}} \times \cancel{11}$$

$$3x = 686$$

$$x = \frac{686}{3}$$

$$= 228.$$

$$\frac{25x}{22} + y = \frac{686}{22}$$

$$\frac{25x}{20} + y = \frac{580}{20}$$